

Aurelio Barrios

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EDUCATION

University of California, San Diego - Halicioğlu Data Science Institute

Bachelor of Science, Data Science with Concentration in Machine Learning

Google x Kura Labs - Agentic Engineer Training Program

Aug 2025 - Oct 2025

- Designed and deployed agentic systems that leveraged task decomposition frameworks, optimizing Agent2Agent communication and enabling scalable, production-ready task orchestration across diverse use cases.
- Developed Model Context Protocol (MCP) server infrastructure for multi-agent systems, applying performance optimizations, role-based authentication, and authorization protocols using LangChain, LangGraph & LangSmith

PROFESSIONAL EXPERIENCE

Learning Engineer, Playlab AI | Remote

May 2025 - Aug 2026

- Engineered an end-to-end automated evaluation pipeline for LLM selection, integrating accuracy, latency and alignment metrics to optimize performance across EdTech applications, improving model selection accuracy by **30%**.
- Implemented scalable data pipelines that ingested, cleaned and transformed large datasets into model-ready formats, reducing preprocessing time by **35%**, improving overall model training efficiency.
- Designed and deployed optimized LLM workflows that balanced latency, cost and accuracy; achieving a **25%** reduction in expenses, while ensuring reliable, production ready AI solutions across multiple EdTech use cases.
- Developed a model orchestration system that dynamically routed requests to the most appropriate LLM, improving accuracy by **30%** while reducing latency and compute costs.

Data Scientist, The Knowledge House | Remote

Jul 2024 - Nov 2025

- Automated the data ingestion process for a high volume API connection handling thousands of user queries for a non-profit per day using ETL/ELT processes and relayed relevant information using Power BI.
- Led core AI infrastructure decisions, such as relational database design, scalable pipeline design and deployment design that supported strategic project goals and secured a **\$100,000** Amazon Web Services AI grant.
- Engineered and deployed the scalable ML pipelines that ingested and processed **years** of learning curriculum and delivered it to a production ready RAG system supporting **hundreds** of queries daily.
- Designed, implemented and deployed a scalable RAG system with semantic search and knowledge graph integration, serving **thousands** of students daily, enhancing personalized learning paths increasing student retention by **60%**.
- Developed model fine-tuning pipelines using Low-Rank Adaptation (LoRA), adapting pre-trained LLMs to EdTech curriculum data, improving domain-specific question and answer performance for **thousands** of students.
- Automated training, versioning and deployment for a RAG system via CI/CD pipelines, ensuring consistent updates, improving reproducibility and supporting model experimentation for different use cases.

Data Science Apprentice, The Knowledge House | Remote

Jul 2023 - Jul 2024

- Led comprehensive training sessions for **10+** fellows on data cleaning and machine learning model optimization in Python and SQL; enhanced their dataset management skills increasing data integrity across multiple data sources, resulting in a **35%** increase in analytical accuracy across multiple machine learning models in our pipeline.
- Built and optimized scalable AWS Glue ELT/ETL pipelines to clean and unify multiple data sources, improving schema management, dataset integrity and ensuring reliable downstream machine learning model performance.

SKILLS

Languages & Tools: Python, SQL, Tableau, PowerBI, Excel

Packages: PyTorch, TensorFlow, LangChain, LangGraph, Pandas, NumPy, PySpark, Boto3

Model Deployment: Docker, Kubernetes, Git, Git Actions, CI/CD, PostgreSQL, YAML

Cloud Computing: Amazon Web Services, Google Cloud Platform, AWS Bedrock

Artificial Intelligence: Retrieval Augmented Generation, Model Context Protocol, Agentic Engineering & Design